

Notes: Cohen, Frazzini, & Malloy (JF, 2010)

Sell-Side School Ties

Contents

1	Big picture	3
2	Data and measurement (school ties, recommendations, and returns)	4
2.1	Why sell-side analysts are the right laboratory	4
2.2	Analyst recommendations and analyst education	4
2.3	Corporate officers, directors, and school information	4
2.4	Final matched sample	5
2.5	Main school-tie variables	5
2.6	Recommendation classification	5
2.7	Portfolio construction and return adjustment	6
2.8	Descriptive facts about schools and coverage	6
3	Models and empirical specifications (all equations plus interpretation)	7
3.1	Defining the school-tie indicators	7
3.2	Calendar-time portfolio test	7
3.3	Panel regressions for recommendation returns	8
3.4	Control variables	8
3.5	Regulation FD interaction regression	9
3.6	U.K. comparison	9
3.7	Strength of ties and alternative social proximity	9
4	Identification and empirical design (what makes the estimates credible)	9
4.1	Why the school-tie comparison is informative	10
4.2	Why this is not just school quality or analyst skill	10
4.3	Why Regulation FD is central	10
4.4	Why the U.K. evidence matters	10
4.5	Interpreting the absence of a sell-side effect	11
4.6	Data-quality and measurement concerns	11
4.7	What the paper can and cannot distinguish	11
5	Tables and figures	12
5.1	Figure 1: XYZ Corp. example	12
5.2	Table I: summary statistics	13

5.3	Table II: school-tie concentration by institution	13
5.4	Table III: returns to school ties, 1993–2006	13
5.5	Table IV: school-tie regressions for buy recommendations	14
5.6	Table V: school-tie regressions for sell recommendations	16
5.7	Table VI: Regulation FD and strength of links	16
5.8	Table VII: U.S. and U.K. Regulation FD evidence	17
5.9	Additional mechanism evidence	18
6	Short summary	18
7	Conclusion	18
7.1	Core conclusion	18
7.2	Economic intuition	19
7.3	Takeaway	19

1 Big picture

- **Question.** Do social networks give sell-side analysts an information advantage when they make stock recommendations?
- **Specific network.** The paper studies educational ties between equity analysts and senior corporate officers or directors. A tie exists when the analyst and a manager/director attended the same academic institution.
- **Core hypothesis.** Analysts with school ties to a firm's management can obtain or interpret firm-specific information better than analysts without such ties. If so, their recommendations should predict higher future stock returns.
- **Setting.** The sample covers U.S. sell-side analyst recommendations from 1993 to 2006, matched to hand-collected analyst education data and BoardEx data on corporate officers and directors.
- **Main empirical design.** The authors compare the performance of recommendations made by linked analysts with recommendations made by nonlinked analysts. They use both calendar-time recommendation portfolios and panel regressions with rich controls and fixed effects.
- **Main return result.** Buy recommendations on school-tied stocks outperform buy recommendations on non-school-tied stocks by about **45–55 basis points per month**, or roughly **5.4–6.6% per year**.
- **Main abnormal-return result.** The school-tie premium remains after Daniel-Grinblatt-Titman-Wermers characteristic adjustment for size, book-to-market, and momentum. Linked buy recommendations have large abnormal returns, while nonlinked buy recommendations have approximately zero abnormal returns.
- **Sell-side asymmetry.** The paper does not find a comparable advantage for sell recommendations. This suggests either that managers mainly reveal positive information or that analysts receive negative information but have weaker incentives to act on it.
- **Main credibility check.** The effect is not just a story about smart analysts from elite schools. It survives controls for Ivy League schools, highly connected schools, highly ranked schools, broker size, analyst experience, all-star status, industry fixed effects, and analyst fixed effects.
- **Regulation FD mechanism test.** Before Regulation FD, the school-tie premium for buy recommendations is around **9.36% per year**. After Regulation FD, the premium is close to zero and statistically insignificant.
- **Out-of-sample comparison.** In the United Kingdom, where the disclosure environment did not experience the same Regulation FD shock during the sample period, the school-tie premium remains large throughout the sample.
- **Economic takeaway.** Informal social networks can be a channel through which private information enters security prices. Regulation aimed at selective disclosure can reduce this informational advantage.

2 Data and measurement (school ties, recommendations, and returns)

2.1 Why sell-side analysts are the right laboratory

- Sell-side analysts produce earnings forecasts, research reports, and stock recommendations.
- Their job requires information gathering, interpretation, and communication to investors.
- If networks matter for information acquisition, analyst recommendations provide a direct market-based outcome: subsequent stock returns.
- Educational ties are attractive empirically because the school connection is formed long before the analyst recommendation and before the firm-specific information being transmitted.

Interpretation. The paper uses analysts as a setting where private information, social access, and market prices can be connected in a measurable way.

2.2 Analyst recommendations and analyst education

- The starting point is the I/B/E/S recommendation database for all sell-side analysts issuing at least one recommendation on a U.S. domestic stock between 1993 and 2006.
- I/B/E/S gives analyst identifiers, analyst names, brokerage identifiers, and recommendation codes.
- The authors remove cases with ambiguous analyst identities, multiple names for one identifier, multiple identifiers for one name, and analyst teams for which full first-name identification is not possible.
- This creates an initial list of **8,620** analysts.
- Analyst education is hand collected from Zoominfo, FINRA BrokerCheck, and other web sources.
- The authors use a conservative matching rule: if an analyst cannot be uniquely matched to an educational background, the observation is discarded.

Interpretation. The main data contribution is not only matching analysts to firms; it is identifying analysts' schools carefully enough to define a credible social-network link.

2.3 Corporate officers, directors, and school information

- Corporate officer and director biographical data come from BoardEx of Management Diagnostics Limited.
- The senior officers are the CEO, CFO, and Chairman.
- The BoardEx data include current and past roles, start and end years, board membership, school attended, and degree type when available.

- The authors hand match academic institutions from the analyst data and BoardEx data to common institution identifiers.
- They also collect school characteristics such as U.S. News rankings, location, and athletic conference.

Interpretation. The paper can observe whether an analyst and a firm insider share an educational background, and can distinguish links to senior management from broader links to the board.

2.4 Final matched sample

- The final sample contains **1,820** analysts.
- These analysts issue **56,994** recommendations.
- The recommendations cover **5,132** CRSP stocks.
- The sample period runs from October 30, 1993 to December 20, 2006.
- On average, the annual sample contains **604** analysts, **1,705** stocks, and **5,746** recommendations.
- The matched sample represents about **23%** of I/B/E/S analysts, **49%** of stocks, **74%** of market equity, and **23%** of recommendations.

Interpretation. The sample is selective because it requires observable education data, but it covers a large share of market capitalization and a meaningful share of the analyst-recommendation universe.

2.5 Main school-tie variables

The paper uses several related link measures.

- **Linked to management:** equals one if the analyst attended the same school as a senior officer of the covered firm.
- **Linked to either management or board:** equals one if the analyst attended the same school as either a senior officer or a director.
- **Fraction linked to analyst:** the fraction of the board or top management connected to the analyst through school ties.
- **Linked by conference:** equals one if the analyst's school and a senior manager's school compete in the same athletic conference, even if they are not the same school.

Interpretation. The first two variables measure direct alumni ties. The fraction-linked variable measures tie strength. The conference variable checks whether weaker social proximity also matters.

2.6 Recommendation classification

- The paper uses the I/B/E/S numeric recommendation code.

- A **BUY** portfolio includes stocks upgraded relative to the prior recommendation, or stocks for which coverage is initiated, resumed, or reiterated with a buy or strong-buy rating.
- A **SELL** portfolio includes stocks downgraded relative to the prior recommendation, stocks initiated/resumed/reiterated with hold, sell, or strong-sell ratings, and stocks dropped from coverage.
- The authors also examine narrower portfolios using only upgrades or only downgrades.
- If a recommendation is not revised or reiterated within 12 months, it expires.

Interpretation. The main comparison is not “buy versus sell.” It is “linked buy versus nonlinked buy” and “linked sell versus nonlinked sell.” This helps isolate the incremental value of the school tie conditional on the recommendation signal.

2.7 Portfolio construction and return adjustment

- For each recommendation, the authors skip one trading day before investing. This avoids using recommendation-day price reactions that may not have been tradable.
- In the buy portfolio, the recommended stock is bought at the close of day $t + 1$ and held until downgrade, dropped coverage, or recommendation expiration.
- In the sell portfolio, stocks are treated analogously on the bearish side.
- Portfolio weights reflect the strength of the recommendation code. For buys, strong buys receive higher weights. For sells, stronger sell signals receive higher weights.
- The paper reports both raw calendar-time returns and characteristic-adjusted abnormal returns.
- Abnormal returns use the Daniel et al. benchmark adjustment based on size, industry-adjusted book-to-market, and one-year momentum quintiles.

Interpretation. The portfolio tests mimic a feasible strategy of following analysts after a one-day delay, and then ask whether school-tied recommendations earn more than otherwise similar non-school-tied recommendations.

2.8 Descriptive facts about schools and coverage

- The mean number of analysts covering a firm in the sample is about **4.97**.
- Sample firms are large and growth-oriented: the average size percentile is **0.78**, while the average book-to-market percentile is **0.37**.
- Harvard University is the largest source of school ties: it accounts for about **18.53%** of analyst ties to senior officials and **18.2%** of analyst ties to corporate boards.
- Ivy League schools account for about **43.7%** of analyst ties to senior officials and **48.5%** of analyst ties to corporate boards.

Interpretation. School links are concentrated among elite institutions, so the paper must show that the results are not merely an elite-school or analyst-quality effect. This is why the later controls and analyst fixed effects are important.

3 Models and empirical specifications (all equations plus interpretation)

3.1 Defining the school-tie indicators

For analyst a , firm j , and year t , the main management-link indicator can be written as

$$\text{MgmtLink}_{ajt} = \mathbf{1} \left\{ \mathcal{S}_a \cap \mathcal{S}_{jt}^{\text{Mgmt}} \neq \emptyset \right\}, \quad (1)$$

where $\mathcal{S}_a$ is the set of schools attended by analyst a , and $\mathcal{S}_{jt}^{\text{Mgmt}}$ is the set of schools attended by firm j 's senior officers in year t .

The broader link variable is

$$\text{EitherLink}_{ajt} = \mathbf{1} \left\{ \mathcal{S}_a \cap \left(\mathcal{S}_{jt}^{\text{Mgmt}} \cup \mathcal{S}_{jt}^{\text{Board}} \right) \neq \emptyset \right\}. \quad (2)$$

Interpretation. A linked recommendation is one where the analyst and at least one firm insider share an alumni network. The management link is narrower and closer to the potential information source; the management-or-board link is broader and captures more possible network paths.

3.2 Calendar-time portfolio test

Let $R_t^{L,B}$ denote the month- t return on the portfolio of linked buy recommendations, and let $R_t^{U,B}$ denote the month- t return on the portfolio of unlinked buy recommendations. The main buy-side spread is

$$LS_t^B = R_t^{L,B} - R_t^{U,B}. \quad (3)$$

Analogously, for sell recommendations,

$$LS_t^S = R_t^{L,S} - R_t^{U,S}. \quad (4)$$

The paper tests whether

$$E[LS_t^B] > 0 \quad (5)$$

for buy recommendations and whether a corresponding linked advantage exists for sell recommendations.

Interpretation. The long-short portfolio holds the recommendation signal fixed. It buys linked buys and shorts nonlinked buys, so the spread is intended to isolate the value of the social tie rather than the value of analysts' buy recommendations in general.

3.3 Panel regressions for recommendation returns

The paper also estimates regressions of future returns on school-tie indicators and controls. A simplified version is

$$r_{aj\tau} = \alpha + \beta \text{Link}_{ajt_0} + X'_{ajt_0} \gamma + \delta_m + \eta_k + \lambda_a + \varepsilon_{aj\tau}, \quad (6)$$

where t_0 is the recommendation date and τ indexes future return observations after the recommendation.

- $r_{aj\tau}$ is the future stock return after analyst a 's recommendation on firm j .
- Link_{ajt_0} is either *Linked to Management* or *Linked to Either*.
- X_{ajt_0} includes firm, analyst, broker, and school controls.
- δ_m denotes recommendation-month fixed effects.
- η_k denotes industry fixed effects where included.
- λ_a denotes analyst fixed effects where included.
- Standard errors are clustered at the recommendation-month level.
- The regressions are run daily, but coefficients are rescaled to represent monthly returns in percent.

Interpretation. The coefficient β asks whether the same type of recommendation earns higher future returns when the analyst has a school tie to the firm. Analyst fixed effects are especially important because they compare linked and nonlinked recommendations made by the same analyst.

3.4 Control variables

The main controls include

- indicators for whether the analyst attended a top-10 most-linked school, an Ivy League school, or a top-40 U.S. News school;
- firm size, book-to-market, and past one-year returns;
- analyst experience;
- an affiliation indicator for underwriting relationships between the analyst's bank and the covered firm;
- Institutional Investor all-star analyst status;
- brokerage size; and
- month, industry, and analyst fixed effects in different specifications.

Interpretation. These controls address the concern that school-tied analysts are simply more talented, better resourced, employed by better brokers, or concentrated among better firms.

3.5 Regulation FD interaction regression

To study the mechanism, the paper exploits the introduction of Regulation FD in October 2000. A simplified regression is

$$r_{aj\tau} = \alpha + \beta_1 \text{MgmtLink}_{ajt_0} + \beta_2 \text{PostFD}_{t_0} + \beta_3 (\text{MgmtLink}_{ajt_0} \times \text{PostFD}_{t_0}) + X'_{ajt_0} \gamma + FE + \varepsilon_{aj\tau}. \quad (7)$$

- β_1 measures the school-tie premium before Regulation FD.
- β_3 measures how much the school-tie premium changes after Regulation FD.
- $\beta_1 + \beta_3$ is the post-Reg FD school-tie premium.

Interpretation. If the school-tie premium comes from selective disclosure, then it should fall after Regulation FD. If the premium comes only from stable ability to assess managerial quality, Regulation FD should not eliminate it.

3.6 U.K. comparison

The paper repeats the portfolio exercise using U.K.-listed stocks.

$$LS_t^{B,UK} = R_t^{L,B,UK} - R_t^{U,B,UK}. \quad (8)$$

Interpretation. The U.K. analysis is an out-of-sample comparison. If the U.S. decline after Regulation FD is really due to the disclosure rule, then a market without a comparable change should not show the same collapse in the school-tie premium.

3.7 Strength of ties and alternative social proximity

The paper also estimates regressions of the form

$$r_{aj\tau} = \alpha + \beta \text{MgmtLink}_{ajt_0} + \phi \text{FracLink}_{ajt_0} + \psi \text{ConferenceLink}_{ajt_0} + X'_{ajt_0} \gamma + FE + \varepsilon_{aj\tau}. \quad (9)$$

- FracLink measures how many firm insiders are linked to the analyst.
- ConferenceLink captures whether the analyst's school and a manager's school share an athletic conference.

Interpretation. If social networks are the channel, then stronger links and weaker-but-related social proximity should also predict performance. The paper finds evidence consistent with this idea.

4 Identification and empirical design (what makes the estimates credible)

4.1 Why the school-tie comparison is informative

- School ties are formed before the analyst produces the recommendation and before the specific firm news being predicted.
- The paper compares recommendations with the same broad signal: linked buys versus nonlinked buys, and linked sells versus nonlinked sells.
- The strongest specifications include analyst fixed effects, so identification comes from the same analyst performing differently on linked versus nonlinked stocks.

Economic message. The core result is not that analysts with elite networks are better overall. It is that analysts perform better on the subset of stocks where they have a specific alumni link.

4.2 Why this is not just school quality or analyst skill

- The paper explicitly controls for attendance at Ivy League schools, highly ranked schools, and schools with many links.
- The school-tie coefficient remains large and statistically significant after these controls.
- Analyst fixed effects remove time-invariant analyst ability, reputation, education, and style.
- Broker size, all-star status, affiliation, and experience controls address the concern that linked analysts are simply better resourced or higher status.

Economic message. The social-network interpretation becomes more credible because the effect is tied to the analyst-firm pair, not simply to analyst quality.

4.3 Why Regulation FD is central

- Regulation FD was designed to reduce selective disclosure from firms to favored market participants.
- If school ties matter because they provide access to selective disclosure, the school-tie premium should decline after Regulation FD.
- The paper finds exactly this: a large pre-Reg FD premium and a near-zero post-Reg FD premium.

Economic message. Regulation FD acts like a mechanism test. It does not prove the exact conversation through which information moved, but it strongly supports the interpretation that school ties were a privileged information channel.

4.4 Why the U.K. evidence matters

- A simple pre/post U.S. test could be confounded by other changes in analyst research markets around 2000.

- The U.K. sample provides a comparison setting without the same sharp disclosure-regulation change.
- The school-tie premium remains large in the U.K. over the full period and does not collapse after 2000.

Economic message. The U.K. result strengthens the claim that the U.S. post-2000 decline is connected to Regulation FD rather than a general global disappearance of analyst-network advantages.

4.5 Interpreting the absence of a sell-side effect

- The buy-side school-tie premium is large and robust.
- The sell-side effect is small and generally insignificant.
- One interpretation is that managers are more willing to reveal positive than negative information.
- Another interpretation is that analysts may learn both positive and negative information but are less willing to publish or act strongly on negative information because of business relationships or implicit norms.

Economic message. The asymmetry is consistent with the institutional reality of sell-side research: optimism, access, and firm relationships can shape how information is used.

4.6 Data-quality and measurement concerns

- Educational backgrounds are hand collected and incomplete for some analysts.
- Graduation years are missing for many observations, so the paper cannot always verify whether the analyst and manager overlapped at the school at the same time.
- A shared school does not guarantee an active personal relationship.
- Conversely, the data miss many real social links not based on education.
- The matched sample is not the full I/B/E/S universe.

Economic message. These limitations mainly create measurement error in the link variable. If many true links are missed or many measured links are weak, the estimated effect may understate the value of active social ties.

4.7 What the paper can and cannot distinguish

- The paper can show that linked analysts' buy recommendations predict higher returns.
- It can show that this advantage weakens sharply after a rule designed to restrict selective disclosure.

- It cannot observe the private conversations or exact pieces of information transmitted.
- It cannot fully distinguish access to material information from better interpretation of soft information, except through the Regulation FD and U.K. evidence.

Economic message. The paper’s strongest claim is about the existence and market value of social-network information advantages, not about the exact micro-level communication technology.

5 Tables and figures

5.1 Figure 1: XYZ Corp. example

Read:

- The figure presents a masked example from the sample.
- A linked analyst upgrades XYZ Corp. to buy, while an unlinked analyst maintains a hold rating.
- After the linked analyst’s upgrade, the firm reports unexpectedly strong earnings and later receives an acquisition offer.
- The stock rises sharply after the linked recommendation.

Economic message. The example illustrates the paper’s core idea: a school-tied analyst appears to make a timely positive recommendation before value-relevant news becomes public.

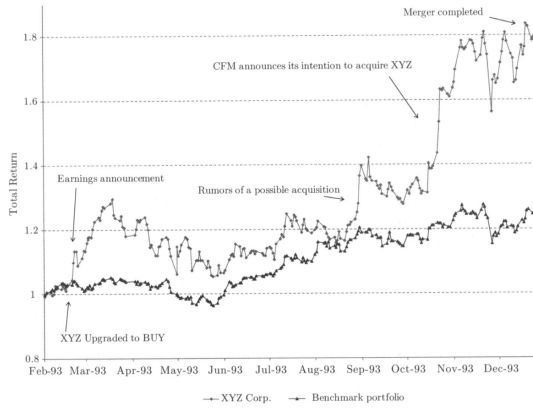


Figure 1. XYZ Corp. This figure shows actual total returns of XYZ Corp. (anonymous name and dates) around the upgrade by a linked analyst, as well as the return on its corresponding risk-adjusted benchmark.

Table I
Summary Statistics

This table reports summary statistics for the sample of sell-side analysts and their covered stocks between 1993 and 2006. The sample of analysts includes all sell-side analysts from merged CRSP, I/B/E/S, BoardEx data that issue recommendations on U.S. stocks between 1993 and 2006. The sample of stocks includes the stocks from merged CRSP, I/B/E/S, BoardEx data with nonmissing information on the educational background of members of the board of directors and senior officers of the firm (CEO, CFO, or Chairman). Panel A reports the data coverage as a fraction of the total number of I/B/E/S analysts, the total number of stocks, the total market value of covered stocks (ME), and the total number of I/B/E/S recommendations (Recs). Panel B reports pooled means. Analyst coverage is the number of analysts providing recommendations for a given stock in the prior 12 months.

Panel A. Coverage of I/B/E/S/CRSP Universe							
Year	No. of Analysts	No. of Stocks	No. of Recs	Fraction of Analysts	Fraction of Stocks	Fraction of ME	Fraction of Recs
1993	153	650	1,066	0.14	0.22	0.52	0.10
1994	243	883	2,468	0.15	0.25	0.54	0.12
1995	283	1,022	2,701	0.16	0.28	0.56	0.12
1996	349	1,166	2,785	0.17	0.28	0.55	0.13
1997	402	1,396	3,339	0.17	0.33	0.66	0.15
1998	516	1,574	4,104	0.19	0.37	0.72	0.16
1999	602	1,737	4,897	0.21	0.44	0.75	0.19
2000	645	1,915	5,562	0.23	0.52	0.84	0.24
2001	682	1,905	6,397	0.25	0.61	0.86	0.28
2002	756	2,203	10,218	0.27	0.68	0.90	0.30
2003	813	2,167	8,829	0.30	0.71	0.90	0.33
2004	958	2,340	9,081	0.33	0.73	0.86	0.36
2005	1,078	2,474	9,374	0.36	0.76	0.88	0.40
2006	971	2,441	9,623	0.33	0.74	0.88	0.38
Average	604	1,705	5,746	0.23	0.49	0.74	0.23

Panel B. Pooled Observations					
	Mean	Median	Min	Max	Std
Analyst coverage per firm	4.97	4.00	1.00	32.00	3.84
Size percentile	0.78	0.84	0.01	1.00	0.20
Book-to-market percentile	0.37	0.33	0.01	1.00	0.25
12-month return percentile	0.52	0.53	0.01	1.00	0.29
No. of schools per year	766	766	707	796	28
No. of board members per year	8,388	8,160	2,355	14,389	4,176
No. of senior officers per year	3,769	3,963	1,183	5,832	1,624

Table II Links between Sell-Side Analysts and Firm Management by Academic Institution							
This table shows summary statistics of the ties among sell-side analysts and U.S. traded firms based on educational backgrounds between 1993 and 2006. The sample of analysts includes all sell-side analysts from merged CRSP, I/B/E/S, and BoardEx data that issue recommendations on U.S. stocks between 1993 and 2006. The sample of stocks includes the stocks from merged CRSP, I/B/E/S, and BoardEx data with nonmissing information on the educational background of members of the board of directors and senior officers of the firm (CEO, CFO, or Chairman). In Panel A we classify a stock as having an educational tie to the analyst if the analyst attended the same institution as a senior officer (defined as either the CEO, CFO, or Chairman of the board). In Panel B we classify a stock as having an educational tie to the analyst if the analyst attended the same institution as a member of the board of directors. The table reports the distribution of the total number of educational links between 1993 and 2006 by academic institution.							
Panel A. Analyst Tied to Firm's Senior Officers				Panel B. Analyst Tied to Board of Directors			
Rank	Academic Institution	No. of Ties	% of Total	Rank	Academic Institution	No. of Ties	% of Total
1	Harvard University	941	18.53	1	Harvard University	2,300	18.22
2	University of Pennsylvania	522	10.28	2	Columbia University	1,139	9.02
3	New York University	350	6.89	3	University of Pennsylvania	1,065	8.44
4	Stanford University	311	6.12	4	New York University	1,006	7.97
5	Columbia University	288	5.67	5	Yale University	717	5.68
6	Cornell University	173	3.41	6	Stanford University	597	4.73
7	M.I.T.	168	3.31	7	M.I.T.	491	3.89
8	Yale University	155	3.05	8	Cornell University	437	3.46
9	University of Chicago	140	2.76	9	UC Berkeley	347	2.75
10	UT Austin	137	2.7	10	University of Chicago	317	2.51
Others		1,893	37.28	Others		4,205	33.32
Ivy League		2,220	43.72	Ivy League		6,122	48.51
All		5,078	100	All		12,621	100

5.2 Table I: summary statistics

Read:

- The matched sample averages 604 analysts, 1,705 stocks, and 5,746 recommendations per year.
- It covers 23% of I/B/E/S analysts, 49% of stocks, 74% of market equity, and 23% of recommendations.
- The mean number of analysts per firm is 4.97.
- Covered firms are large: their average size percentile is 0.78.
- Covered firms are also more growth-oriented: their average book-to-market percentile is 0.37.

Economic message. The paper's sample is not the whole market, but it covers the economically important part of the market where analyst coverage and institutional attention are high.

5.3 Table II: school-tie concentration by institution

Read:

- Table II reports the distribution of school ties by academic institution.
- Harvard is the largest source of analyst-management and analyst-board ties.
- Ivy League schools account for a large share of the observed links: about 43.7% of analyst ties to senior officials and 48.5% of analyst ties to boards.

Economic message. Because ties are concentrated in elite schools, the later controls for school quality and school connectedness are essential. The results would be less convincing if they only reflected Harvard or Ivy League status.

5.4 Table III: returns to school ties, 1993–2006

Read:

- For buy recommendations, nonlinked recommendations earn 1.04% per month in raw returns.

Table III
Returns to School Ties, 1993–2006

This table shows calendar time portfolio returns. We classify a stock as having an educational tie to the analyst if the analyst attended the same institution as a senior officer (CEO, CFO, or Chairman) or a board member. Each recommendation is assigned to one of two portfolios: (1) a BUY portfolio consisting of all stocks upgraded with respect to the previous recommendation, or for which coverage is initiated, resumed, or reiterated with a buy (I/B/E/S code = 2) or strong buy (I/B/E/S code = 1) rating; and (2) a SELL portfolio, consisting of all stocks downgraded with respect to the previous recommendation, for which coverage is initiated, resumed, or reiterated with a hold (I/B/E/S code = 3), sell (I/B/E/S code = 4), or strong sell (I/B/E/S code = 5) rating; or that are dropped from coverage. If the brokerage house does not report the stock as dropped from coverage and a recommendation is not revised or reiterated within 12 months, it is considered expired. We skip a trading day between recommendation and investment (disinvestment). For the BUY portfolio each recommended stock is held until it is downgraded or dropped from coverage, or until the recommendation expires. We compute value-weighted portfolios by averaging across analysts, weighting individual recommendations by the I/B/E/S recommendation code; for the BUY portfolio, we reverse these recommendation codes so that a strong buy is set to 5 and a strong sell is set to 1. The SELL portfolio is constructed in a similar fashion with the exception that the original I/B/E/S recommendation codes (i.e., strong sell = 5, strong buy = 1) are used as the portfolio weights. We report average returns and DGTW-adjusted returns for the period 1993 to 2006. DGTW characteristic-adjusted returns are defined as raw returns minus the returns on a value-weighted portfolio of all CRSP firms in the same size, industry-adjusted market-to-book, and 1-year momentum quintile. Returns are in monthly percent. L/S is the average return of a zero cost portfolio that holds the portfolio of linked stocks and sells short the portfolio of nonlinked stocks. *t*-statistics are shown below the estimates, and 1%, 5%, and 10% statistical significance are indicated with ***, **, and *, respectively.

Panel A: Buy Recommendations					
	Buy Recommendations (Level)		Only Upgrades		
	Raw Returns	Abnormal Returns	Raw Returns	Abnormal Returns	
No shared educational background	1.04** (1.97)	0.04 (0.43)	1.35*** (2.81)	0.31*** (3.14)	
Linked recommendations		L/S		L/S	
Analyst linked to senior management	1.59*** (3.94)	0.55*** (3.75)	0.51*** (3.19)	0.47*** (3.96)	1.65*** (3.16)
Analyst linked to senior management or board of directors	1.49*** (2.91)	0.45*** (3.87)	0.44*** (3.24)	0.40*** (4.63)	1.70*** (3.35)

(continued)

Table III—Continued

Panel B: Sell Recommendations					
	Sell Recommendations (Level)		Only Downgrades		
	Raw Returns	Abnormal Returns	Raw Returns	Abnormal Returns	
No shared educational background	1.03* (1.83)	−0.17 (−1.31)	1.06* (1.81)	−0.21 (−1.45)	
Linked recommendations		L/S		L/S	
Analyst linked to senior management	1.10** (2.05)	0.07 (0.46)	0.10 (0.61)	0.27* (1.92)	1.36** (2.20)
Analyst linked to senior management or board of directors	1.08** (2.08)	0.05 (0.49)	0.05 (0.37)	0.22** (2.26)	1.09** (2.05)

- Buy recommendations linked to senior management earn 1.59% per month; the linked-minus-nonlinked spread is **0.55% per month**.
- The corresponding abnormal-return spread is **0.47% per month**.
- When links to either senior management or the board are used, the raw spread is **0.45% per month** and the abnormal spread is **0.40% per month**.
- Using only upgrades, the linked-minus-nonlinked spread remains positive, around 0.29–0.35% per month depending on the link definition and return adjustment.
- For sell recommendations, linked and nonlinked returns are not meaningfully different in the way the buy-side results are.

Economic message. The central empirical result is visible before any regression: linked buy recommendations outperform nonlinked buy recommendations by economically large amounts.

5.5 Table IV: school-tie regressions for buy recommendations

Read:

- The coefficient on *Linked to Management* is about **0.48% per month** in the simplest specification.
- The coefficient on *Linked to Either* is about **0.37% per month**.
- The result remains significant after controls for top linked schools, Ivy League schools, top-40 U.S. News schools, firm characteristics, analyst characteristics, broker characteristics, and fixed effects.
- With analyst fixed effects, the management-link coefficient remains positive and significant, around **0.32% per month** in the tightest specification.

Economic message. The school-tie premium is not explained by elite-school attendance, general analyst ability, firm type, broker resources, or analyst status.

Table IV
School-Tie Regressions for Buy Recommendations

This table reports panel regressions of returns on buy recommendations of analysts. The dependent variable is future returns. The regressions were run daily, but coefficients have been adjusted to represent monthly returns in percent. The first two variables are categorical variables indicating whether or not the analyst is linked to the given firm on which she is making a recommendation via an education network: (i) *Linked to Mgmt* indicates that the analyst is linked to the senior officers, and (ii) *Linked to Either* indicates that the analyst is linked to either the senior officers or the board of directors. *Top 10 Most Linked* is a categorical variable indicating whether an analyst attended a school with the highest number of links to senior management (or the board of directors) in our sample. *Ivy League* is a categorical variable indicating whether an analyst attended a school in the Ivy League. *Top 40 U.S. News* is a categorical variable indicating whether an analyst attended a school ranked in the Top 40 National Universities in *U.S. News and World Report*. *Size* is the market capitalization of the firm, *BM* is the book-to-market ratio of the firm, and *Past Returns* is the past 1-year stock return of the firm. *Analyst Experience* is equal to the number of years the analyst has been making recommendations recorded in I/B/E/S. *Affiliation* is a categorical variable that measures whether the given firm has an underwriting relationship with the analyst's brokerage firm. *All-Star* is a categorical variable equal to one if the investor was voted an all-star analyst in the October issue of *Institutional Investor* magazine for the given year. *Brokerage Size* is the total number of analysts that work at the given analyst's brokerage house. Fixed effects for month (Month), analyst (Analyst), and industry (Indus) using Fama-French industry definitions are included where indicated. All standard errors are adjusted for clustering at the month level. *t*-statistics are shown below the coefficient estimates, and 1%, 5%, and 10% statistical significance are indicated with ***, **, and *, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Linked to Mgmt	0.48*** (4.23)		0.43*** (3.66)	0.44*** (3.77)	0.43*** (3.61)	0.39*** (3.30)	0.32** (2.51)
Linked to Either		0.37*** (5.16)					
Top 10 Most Linked			0.04 (0.67)				
Ivy League				0.01 (0.18)			
Top 40 U.S. News					0.05 (0.69)		
Size			-0.01*** (4.62)	-0.01*** (4.73)	-0.01*** (4.77)		-0.01*** (5.44)
BM			0.68*** (5.68)	0.67*** (5.71)	0.68*** (5.64)		0.69*** (6.23)
Past Returns			-0.07 (0.78)	-0.07 (0.85)	-0.07 (0.80)		-0.14* (1.84)
Analyst Experience			0.03*** (2.65)	0.03*** (2.62)	0.03*** (2.86)		-0.29*** (2.88)
Affiliation			-0.32* (1.79)	-0.33* (1.72)	-0.32* (1.79)		-0.43** (2.43)
All-Star			-0.02 (0.16)	-0.13 (1.21)	-0.01 (0.14)		0.05 (0.25)
Brokerage Size			0.00*** (2.59)	-0.00** (2.11)	-0.00*** (2.82)		-0.01*** (3.59)
Fixed Effect	Month	Month	Month	Month	Month	Month	Month
Fixed Effect			Indus	Indus	Indus	Analyst	Analyst

5.6 Table V: school-tie regressions for sell recommendations

Read:

- The link coefficients for sell recommendations are small and statistically insignificant.
- In the strictest specifications, linked sell recommendations do not reliably outperform nonlinked sell recommendations.
- Control variables such as size, book-to-market, and past returns matter, but the network effect is not robust on the sell side.

Economic message. The network advantage appears primarily on positive recommendations. This fits the idea that firm insiders are more willing to transmit good news, or that analysts are more willing to act on good news than bad news.

Table V
School-Tie Regressions for Sell Recommendations

This table reports panel regressions of returns on sell recommendations of analysts. The dependent variable is future returns. The regressions were run daily, but coefficients have been adjusted to represent monthly returns in percent. All independent variables are defined as in Table IV. Fixed effects for month (Month), analyst (Analyst), and industry (Indus) using Fama-French industry definitions are included where indicated. All standard errors are adjusted for clustering at the month level. *t*-statistics are shown below the coefficient estimates, and 1%, 5%, and 10% statistical significance are indicated with ***, **, and *, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Linked to Mgmt	0.02 (0.25)		0.04 (0.40)	0.04 (0.34)	0.05 (0.41)	-0.03 (0.34)	-0.06 (0.48)
Linked to Either		0.01 (0.20)					
Top 10 Most Linked			0.01 (0.11)				
Ivy League				-0.01 (0.08)			
Top 40 U.S. News					0.02 (0.27)		
Size			-0.01*** (3.79)	-0.01*** (3.82)	-0.01*** (3.77)		-0.01*** (4.66)
BM			0.45*** (3.93)	0.41*** (3.40)	0.45*** (3.92)		0.48*** (4.45)
Past Returns			-0.28** (2.24)	-0.35*** (2.77)	-0.30** (2.41)		-0.37*** (3.31)
Analyst Experience			0.02 (1.44)	0.02 (1.52)	0.02 (1.34)	-0.22* (1.69)	
Affiliation			-0.23 (0.76)	-0.36 (1.06)	-0.29 (0.95)	-0.27 (0.83)	
All-Star			-0.18 (1.65)	-0.11 (0.92)	-0.18 (1.55)	-0.10 (0.67)	
Brokerage Size			0.00 (0.86)	0.00 (0.75)	0.00 (0.94)	0.00 (0.75)	
Fixed Effect	Month	Month	Month	Month	Month	Month	Month
Fixed Effect			Indus	Indus	Indus	Analyst	Analyst

Table VI
Regulation FD and Strength of Links

This table reports panel regressions of returns on buy recommendations of analysts. The dependent variable is future returns. The regressions were run daily, but coefficients have been adjusted to represent monthly returns (abnormal returns) in percent. *Linked to Mgmt* is defined as in Table III. *Post-Reg FD* is a categorical variable equal to one for all recommendations made after Regulation FD came into effect (Oct 23, 2000), and zero for all recommendations made before. *Link Mgmt*Post Reg-FD* is the interaction term between *Linked to Mgmt* and *Post-Reg FD*. *Frac Link to Analyst* is the fraction of the board of directors that is linked to the analyst. *Linked by Conf* indicates whether the analyst attended a school competing in the same athletic conference as that of a senior manager. All other independent variables are defined as in Table IV. Column 3 (Sub) includes only the subset of analysts that are in the sample both pre- and post-Reg FD. Fixed effects for month (Month), analyst (Analyst), and industry (Indus) using Fama-French industry definitions are included where indicated. All standard errors are adjusted for clustering at the month level. *t*-statistics are shown below the coefficient estimates, and 1%, 5%, and 10% statistical significance are indicated with ***, **, and *, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Full	Full	Sub	Full	Full	Full	Full
Linked to Mgmt	0.79*** (4.31)	0.72*** (3.61)	0.88*** (3.88)			0.31*** (2.68)	0.31*** (2.59)
Link Mgmt* Post-Reg FD	-0.58** (2.51)	-0.63** (2.53)	-0.91*** (3.03)				
Frac Link to Analyst				0.69*** (2.62)	0.57** (2.07)		
Linked to Either					0.19** (2.43)		
Linked by Conf						0.22*** (2.59)	0.15** (1.98)
Post-Reg FD	-0.97*** (3.68)	-1.78*** (3.43)	-1.82*** (3.56)				
Top 10	0.03 (0.43)					0.02 (0.31)	0.02 (0.27)
Size	-0.01*** (5.24)	-0.01*** (6.55)	-0.01*** (5.89)	-0.01*** (5.39)	-0.01*** (5.47)		-0.01*** (5.26)
Bm	0.62*** (4.61)	0.67*** (5.58)	0.79*** (5.91)	0.70*** (6.24)	0.70*** (6.24)		0.56*** (4.35)
Past returns	-0.19* (1.94)	-0.27*** (3.54)	-0.24*** (3.01)	-0.14* (1.83)	-0.14* (1.83)		-0.06 (0.62)
Analyst Experience	0.03** (2.16)	0.16*** (2.74)	0.17*** (3.03)	-0.29*** (2.85)	-0.29*** (2.85)		0.03*** (2.75)
Affiliation	-0.83*** (3.73)	-0.83*** (4.31)	-0.80*** (3.54)	-0.42** (2.40)	-0.42** (2.40)		-0.31* (1.72)
All-Star	-0.04 (0.31)	0.17 (0.78)	0.17 (0.74)	0.05 (0.25)	0.05 (0.24)		-0.07 (0.71)
Brokerage Size	-0.00** (2.30)	-0.01*** (4.04)	-0.01*** (4.17)	-0.01*** (3.61)	-0.01*** (3.62)		-0.00*** (2.96)
Fixed Effect	Indus	Analyst	Analyst	Analyst	Analyst	Month	Month
Fixed Effect		Indus	Indus	Month	Month		

5.7 Table VI: Regulation FD and strength of links

Read:

- Before Regulation FD, the management-link coefficient is about **0.72% per month**, or roughly **8.64% per year**.
- The interaction between management links and the post-Reg FD period is strongly negative.

- The implied post-Reg FD school-tie premium is only about **0.09% per month** and statistically insignificant.
- Stronger ties matter: the fraction of top firm insiders linked to the analyst has a positive and significant coefficient.
- Weaker social proximity also matters: a same-athletic-conference link produces a positive coefficient, though smaller than the direct school-tie coefficient.

Economic message. The Reg FD result is the paper’s cleanest mechanism evidence. The school-tie premium looks like a selective-disclosure channel that becomes much less valuable once selective disclosure is restricted.

5.8 Table VII: U.S. and U.K. Regulation FD evidence

Table VII
Regulation FD: U.S. and U.K. Evidence

This table shows calendar time portfolio returns (in local currency). We classify a stock as having an educational tie to the analyst if the analyst attended the same institution as a senior officer (CEO, CFO, or Chairman) or a board member. Each recommendation is assigned to one of two portfolios: (1) a BUY portfolio consisting of all stocks upgraded with respect to the previous recommendation, or for which coverage is initiated, resumed, or reiterated with a buy (I/B/E/S code = 2) or strong buy (I/B/E/S code = 1) rating, and (2) a SELL portfolio, consisting of all stocks downgraded with respect to the previous recommendation, for which coverage is initiated, resumed, or reiterated with a hold (I/B/E/S code = 3), sell (I/B/E/S code = 4), or strong sell (I/B/E/S code = 5) rating, or that are dropped from coverage. If the brokerage house does not report the stock as dropped from coverage and a recommendation is not revised or reiterated within 12 months, it is considered expired. We skip a trading day between recommendation and investment (date of investment). For the BUY portfolio each recommended stock is held until it is downgraded or dropped from coverage, or until the recommendation expires. We compute value-weighted portfolios by averaging across analysts, weighting individual recommendations by the I/B/E/S recommendation code; for the SELL portfolio, we reverse these recommendation codes so that a strong buy is set to 5 and a strong sell is set to 1. The SELL portfolio is constructed in a similar fashion with the exception that the original I/B/E/S recommendation codes (i.e., strong sell = 5, strong buy = 1) are used as the portfolio weights. Panel A presents average returns on U.S. stocks for two subperiods, Pre- and Post-Reg FD, corresponding to returns for recommendations issued prior and subsequent to the introduction of Regulation FD in the United States on October 23, 2000. Panel B presents average returns and DGTW-adjusted returns over the period 1993 to 2006 for U.K. stocks; DGTW characteristic-adjusted returns are defined as raw returns minus the returns on a value-weighted portfolio of all I/B/E/S firms traded in the United Kingdom in the same size, (industry-adjusted) market-to-book, and 1-year momentum quintile. Analogous to our U.S. sample, we collect educational data on I/B/E/S analysts issuing recommendations on stocks traded in the United Kingdom, as defined by the I/B/E/S country exchange code. We hand matched firms from the BoardEx sample to I/B/E/S using company names. Daily returns (in local currency) are from Factset. Panel C presents average returns on U.K. stocks for two subperiods, Pre- and Post-Reg FD, corresponding to returns for recommendations issued prior and subsequent to the introduction of Regulation FD in the United States on October 23, 2000. Returns are in monthly percent. L/S is the average return of a zero-cost portfolio that holds the portfolio of linked stocks and sells short the portfolio of nonlinked stocks. t-statistics are shown below the estimates, and 1%, 5%, and 10% statistical significance are indicated with ***, **, and *, respectively.

Panel A. U.S. Pre- and Post-Reg FD									
	Buy Recommendations			Sell Recommendations					
	Pre-Reg FD	Post-Reg FD	Difference	Pre-Reg FD	Post-Reg FD	Difference			
No shared educational background	1.25 (1.93)	0.78 (0.87)	0.50 (0.46)	0.92 (1.57)	1.18 (1.12)	-0.26 (-0.23)	L/S		
Analyst linked to senior management	2.03*** (3.12)	0.78*** (3.20)	1.01 (1.19)	0.51* (0.95)	1.20*** (1.73)	-0.29 (-1.99)	0.97 (1.31)	0.24 (1.00)	0.50 (1.85)
Analyst linked to senior management or board of directors	1.94*** (3.12)	0.68*** (4.36)	0.14 (0.84)	0.55*** (1.01)	1.07* (2.38)	-0.09 (-1.87)	1.09 (1.12)	-0.01 (-0.53)	0.25 (1.13)

(continued)

Table VII—Continued

Panel B. U.K. Sample, 1993–2006

	Buy Recommendations		Sell Recommendations							
	Raw Returns	Abnormal Returns	Raw Returns	Abnormal Returns						
No shared educational background	0.71 (1.41)	L/S (0.51)	-0.13 (0.51)	0.45 (0.78)			-0.24 (0.91)	L/S		
Analyst linked to senior management	2.58*** (3.11)	1.87*** (2.79)	1.54** (2.09)	1.67*** (2.30)	-0.34 (1.06)	-0.79 (0.94)	0.02 (0.04)	0.26 (0.46)		

Panel C. U.K. Pre- and Post-Reg FD

	Buy Recommendations			Sell Recommendations							
	Pre-Reg FD	Post-Reg FD	Difference	Pre-Reg FD	Post-Reg FD	Difference					
No shared educational background	0.54 (0.71)	0.91 (1.47)	-0.37 (-0.36)	0.38 (0.47)	0.27 (0.80)	-0.19 (-0.16)	L/S				
Analyst linked to senior management	2.32** (1.98)	1.78** (2.11)	2.90** (2.48)	1.99* (1.82)	-0.58 (0.34)	-0.21 (0.15)	-0.48 (0.52)	-0.86 (1.18)	-0.08 (0.04)	-0.65 (0.40)	-0.21 (0.23)

Read:

- In the U.S. pre-Reg FD period, the linked-minus-nonlinked buy spread is about **0.78% per month** for management links and **0.68% per month** for management-or-board links.
- In the U.S. post-Reg FD period, the buy spread falls to about **0.26%** or **0.14%** per month and is not statistically significant.
- In the U.K. full-sample test, linked buy recommendations strongly outperform nonlinked buy recommendations: the raw spread is about **1.87% per month**, and the abnormal spread is about **1.67% per month**.
- In the U.K., the linked buy advantage remains large both before and after the U.S. Regulation FD date.
- Sell recommendations again show no comparable linked advantage.

Economic message. The U.S. effect disappears after the regulation, while the U.K. effect persists. This cross-market comparison supports the view that Regulation FD reduced analysts’ private network advantage in the United States.

5.9 Additional mechanism evidence

Read:

- A one-standard-deviation increase in the number of analyst connections increases the probability of becoming an Institutional Investor all-star by nearly 50%, from 9.2% to 13.6%.
- Forecast-accuracy tests do not show strong evidence that linked analysts are significantly better at forecasting earnings per share.
- Merger-related tests do not suggest that takeover information alone explains the school-tie premium.

Economic message. Connections appear valuable for analyst reputation and stock recommendations, but the information being transmitted is not simply earnings-per-share forecasts or merger news.

6 Short summary

- The paper studies whether school ties between sell-side analysts and corporate insiders improve analyst recommendations.
- It matches I/B/E/S recommendations to hand-collected analyst education data and BoardEx officer/director education data.
- Linked buy recommendations outperform nonlinked buy recommendations by about 5.4–6.6% per year.
- The result survives controls for elite schools, highly connected schools, analyst quality, broker characteristics, firm characteristics, and analyst fixed effects.
- The effect is concentrated in buy recommendations, not sell recommendations.
- The premium is large before Regulation FD and nearly zero afterward.
- In the U.K., where the same disclosure-regulation shock did not occur, the school-tie premium remains large.
- Stronger links and broader social proximity are also associated with better performance.
- The paper’s interpretation is that social networks helped analysts obtain private information, and Regulation FD reduced this channel.

7 Conclusion

7.1 Core conclusion

Cohen, Frazzini, and Malloy show that sell-side analysts perform better when they have educational ties to the firms they cover. The strongest evidence comes from buy recommendations: school-tied

buy calls earn substantially higher future returns than non-school-tied buy calls.

7.2 Economic intuition

The economic mechanism is information flow through social networks. Shared schools can lower the cost of access, increase trust, or make communication easier between analysts and firm insiders. Before Regulation FD, this channel appears to have helped analysts produce more profitable buy recommendations. After Regulation FD, the channel becomes much weaker.

7.3 Takeaway

The paper gives market-based evidence that social networks matter for the transmission of private information. It also shows that disclosure regulation can change how information advantages are distributed across market participants.